Behavioral/Cognitive

Anatomically Distinct Regions in the Inferior Frontal Cortex Are Modulated by Task and Reading Skill

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The inferior frontal cortex (IFC) is a critical region for reading and language. This part of the cortex is highly heterogeneous in its structural and functional organization and shows high variability across individuals. Despite decades of research, the relationship between specific IFC regions and reading skill remains unclear. To shed light on the function of IFC in reading, we aim to (1) characterize the functional landscape of text-selective responses in IFC, while accounting for interindividual variability, and (2) examine how text-selective regions in the IFC relate to reading proficiency. To this end, children with a wide range of reading ability ($N = 66$; age 7–14 years, 34 female, 32 male) completed functional MRI scans while performing two tasks on text and nontext visual stimuli. Importantly, both tasks do not explicitly require reading and can be performed on all visual stimuli. This design allows us to tease apart stimulus-driven responses from task-driven responses and examine where in IFC task and stimulus interact. We were able to identify three anatomically distinct, text-selective clusters of activation in the IFC, inferior frontal sulcus (IFS), and dorsal and ventral precentral gyrus (PrG). These three regions showed a strong task effect that was highly specific to text. Furthermore, text selectivity in the IFS and dorsal PrG was associated with reading proficiency, such that better readers showed higher selectivity to text. These findings suggest that text-selective regions in the IFC are sensitive to both stimulus and task and highlight the importance of this region for proficient reading.

Key words: dyslexia; inferior frontal cortex; reading; task effects; text selectivity; visual processing

Significance Statement

The inferior frontal cortex (IFC) is a critical region for language processing, yet despite decades of research, its relationship with reading skill remains unclear. In a group of children with a wide range of reading skills, we were able to identify three anatomically distinct text-selective clusters of activation in the IFC. These regions showed a strong task effect that was highly selective to text. Text selectivity was positively correlated with reading proficiency, such that better readers showed higher selectivity to text, even in tasks that did not require reading. These findings suggest that multiple text-selective regions within IFC are sensitive to both stimulus and task and highlight the critical role of IFC for reading proficiency.

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Introduction

The inferior frontal cortex (IFC) has long been recognized as a critical region for language. This association dates back to the 19th century, with the discovery of an inferior frontal speech production region, now known as Broca's area (Broca, 1861). By the 1990s, new high-resolution imaging methods enabled researchers to further parse functionally specific subregions within IFC, including distinct regions for semantic and phonological processing (Shaywitz et al., 1994; Poldrack et al., 1999; Vigneau et al., 2006). Phonological processing, or the ability to manipulate the sounds of spoken language, is essential for fluent reading and

is often impaired in struggling readers (Bradley and Bryant, 1983; Wagner and Torgesen, 1987). Consequently, neuroimaging studies have focused on IFC as a region where differences between struggling and typical readers may emerge, yet these findings have been inconsistent (Norton et al., 2015). Some have reported reduced IFC activation in struggling readers compared to typical readers (Maisog et al., 2008; Zhang and Peng, 2022), while others found increased IFC activation in struggling readers (Pugh et al., 2001; Hoeft et al., 2007; Shaywitz and Shaywitz, 2008). Surprisingly, the most consistent location where group differences have been reported is not a phonological, but a visual region—the visual word form area (VWFA) in the ventral occipitotemporal cortex (Kubota et al., 2019; Brem et al., 2020). Thus, despite the well-established role of IFC in phonological processing and language, its relationship with reading proficiency remains unclear.

The conflicting findings regarding the role of IFC in reading difficulties may arise from several factors, including variations in the tasks used (McNorgan et al., 2015; Norton et al., 2015), the stimulus and baseline conditions, participants' age (Richlan et al., 2011), the region investigated within IFC, and lack of anatomical precision in defining regions across participants. Intersubject variability can confound studies that transform individual functional data into a standard anatomical space and perform group-wise statistics. This "group average" approach can inadvertently group together different functional regions across participants, potentially averaging out meaningful differences (Glezer and Riesenhuber, 2013). This is especially pertinent in the frontal cortex where anatomical and cytoarchitectural individual differences are prominent (Ruland et al., 2022; Nolan et al., 2024) and functional language regions have high anatomical variability (Fedorenko and Kanwisher, 2009).

An additional complexity of IFC is its role as a high-level regulatory region implicated in working memory, attention, and other high-level, domain-general processes (Miller and Cohen, 2001; Tops and Boksem, 2011). Several studies of vision have shown that activity in IFC is sensitive to both stimulus and task demands (McKee et al., 2014; Bugatus et al., 2017). However, studies of reading in IFC have typically used language tasks, such as sentence reading, where teasing apart the stimulus-driven response from task demands is not straightforward. Thus, it remains unclear whether IFC is sensitive to visual text, the task demands imposed by reading, or their interaction.

To address these questions, here we aim to (1) characterize the functional landscape of text-selective responses in IFC, while accounting for individual variability; (2) distinguish task effects, stimulus effects, and task–stimulus interactions; and (3) examine how text-selective regions in IFC relate to reading skill. We adopted an individualized approach to identify text-selective regions on the cortical surface of children with a wide range of reading abilities. Importantly, our experimental paradigm manipulates task demands while keeping the stimuli and experimental structure constant, allowing us to tease apart stimulus-driven responses from task-driven responses. We use two tasks that do not explicitly require reading: a fixation task where participants make perceptual judgments on a fixation dot while viewing (but ignoring) images of different visual categories, and a one-back task, where they respond when the same image repeats. This task choice is critical to ensure that participants can perform the same task on both text (real words, pseudowords, consonant strings) and nontext (false-fonts, faces, limbs, objects) stimuli and that individual differences are not confounded by differences in task performance.

Materials and Methods

Overview

This study is a subset of a larger, longitudinal study in which participants completed multiple behavioral assessments and MRI scans over the course of 13 months. Here, we report on cross-sectional analyses from the baseline timepoint of the longitudinal study.

Participants

A total of 78 subjects participated in the longitudinal study. For this analysis of the baseline timepoint, we excluded 10 participants due to high motion in the scanner (for exclusion criteria, see below, BOLD response estimation) and two participants who did not complete the scan. Our final sample for analysis included 66 subjects (mean age, 10 ± 1.37 years; range, 7.8–13.8; 34 female, 32 male; Table 1) that had at least one usable run from each task. All subjects had normal or corrected-to-normal vision and no neurological disorders or hearing difficulty. All subjects were either monolingual English speakers or communicated in English at least 60% of the day and had acquired English before the age of 3. Additionally, participants were screened using the vocabulary and the matrix reasoning subtests of the Wechsler Abbreviated Scale of Intelligence, Second Edition (WASI-II; Wechsler, 2011) to ensure their score was above the sixteenth percentile for their age group. WASI scores were not available for four participants. One participant was included in the study despite having a lower WASI score due to the experimenter's assessment that their bilingualism was biasing their score in a way that was not representative of their true ability. All protocols were approved by the Stanford Internal Review Board on Human Subjects Research and informed assent and consent were obtained from each child and guardian, respectively.

Behavioral assessments

Participants completed the Woodcock–Johnson Test of Achievement IV (Schrank and Wendling, 2018). The assessments were administered over Zoom as data collection for this study began during the COVID-19 pandemic. We used the basic reading score (BRS) measure, which is an age-standardized composite of real word reading (letter-word ID subtest) and pseudoword reading (word attack subtest). BRS scores were used in analyses as a continuous variable. For some visualizations, we split the sample into struggling readers and typical readers by a reading score cutoff of BRS <95 (see below, Statistical analysis). Behavioral assessments took place within a couple of weeks of the MRI scan. Assessments were independently scored by two trained researchers, and a third researcher was brought in to adjudicate any discrepancies.

Table 1. Demographic information (*N* = 66)

	N (%)	Mean (SD)	Range
Gender			
Female	34 (51.5)		
Male	32 (48.5)		
Handedness			
Right	57 (86.4)		
Left	7 (10.6)		
Mixed	2 (3.0)		
Grade			
1	1 (1.5)		
2	12 (18.2)		
3	19 (28.8)		
4	17 (25.8)		
5	9 (13.6)		
6	5 (7.6)		
7	1 (1.5)		
8	2 (3)		
Age (years)		10 (1.37)	[7.8, 13.8]
WJ basic reading score		89.27 (13.56)	[63, 126]
WASI full-scale IQ		106.90 (10.42)	[79, 127]

WJ, Woodcock–Johnson; WASI, Wechsler Abbreviated Scale of Intelligence.

Experimental design

The experimental design followed the localizer experiment in White et al. (2023) with some minor adjustments to make it suitable for children. We summarize the details here for convenience.

Stimuli. Subjects were presented with stimuli from five visual categories: text, false-fonts, objects, faces, and limbs (Fig. 1). The text category was composed of four subcategories of strings: high-frequency words, low-frequency words, pronounceable pseudowords, and consonant strings. The false-font category included two different sets of false characters, one matched in low-level features to the Sloan font and one matched in low-level features to the Courier font (Vidal et al., 2017). The stimuli were arranged such that a small image was presented at the center of the screen with a fixation dot (0.14° visual angle wide), and two larger images from the same visual category were presented on either side of fixation to balance the visual acuity and processing advantage of the fovea compared to the peripheral retina.

Each trial presented stimuli for 800 ms followed by 200 ms of a blank fixation screen. Four trials of a single visual category were presented in a row to form a block. A single run of the experiment randomly presented 10 blocks of each visual category, except for the text category which included 20 blocks (5 blocks each of high-frequency words, low-frequency words, pseudowords, and consonant strings).

Tasks. In each run, participants were asked to complete either a one-back task or a fixation task. In the one-back task, participants viewed the stimuli while fixating and responded via a button press when an image repeated twice in a row (Fig. 1*A*). In the fixation task, participants viewed the stimuli while fixating and responded via a button press when the fixation dot changed color (Fig. 1*B*). The one-back task was designed to draw the participant's attention to the visual stimuli while the fixation task was designed to draw the participant's attention away from the visual stimuli to the fixation dot. The original experiment in White et al. (2023) was designed and piloted to maximize participant engagement, equate task difficulty, and avoid ceiling effects. Following pilot data in children which revealed that the experiment was too difficult (yielded an accuracy of <10% and high dropout), we made several adjustments to make it more child-friendly. Specifically, the fixation dot was increased in size and rotated between three easy to differentiate colors (red, yellow, cyan), instead of the six colors in the original version. In addition, the presentation duration was increased, and the interstimulus interval was shortened by 100 ms compared with the original version to alleviate the working memory load. Image repetitions and fixation-dot color changes occurred at random in 33% of trials, independent of each other and of the presented visual category. This design was intended to optimize the child experience while ensuring engagement, but note that it results in very few target trials per category per subject (median = 5) which makes analysis of the behavioral data by category unwarranted. Participants held the response box in their dominant hand and used their index finger to press the response button. Participants completed four runs of the experiment, alternating between the one-back task and the fixation task. Participants were visually monitored during each run, and the entire run was discarded if the participant fell asleep at any point.

MRI

Acquisition. Participants were scanned using a General Electric Sigma MR750 3 T scanner at Stanford University's Center for Cognitive and Neurobiological Imaging. Participants first completed an introduction session where they practiced a short version of the experiment in a scanner simulator to get acclimated to the scanner noises, the tasks, and the response box. During this session, they also practiced staying still with a feedback system (MoTrak Head Motion Tracking System; RRID:SCR_009607).

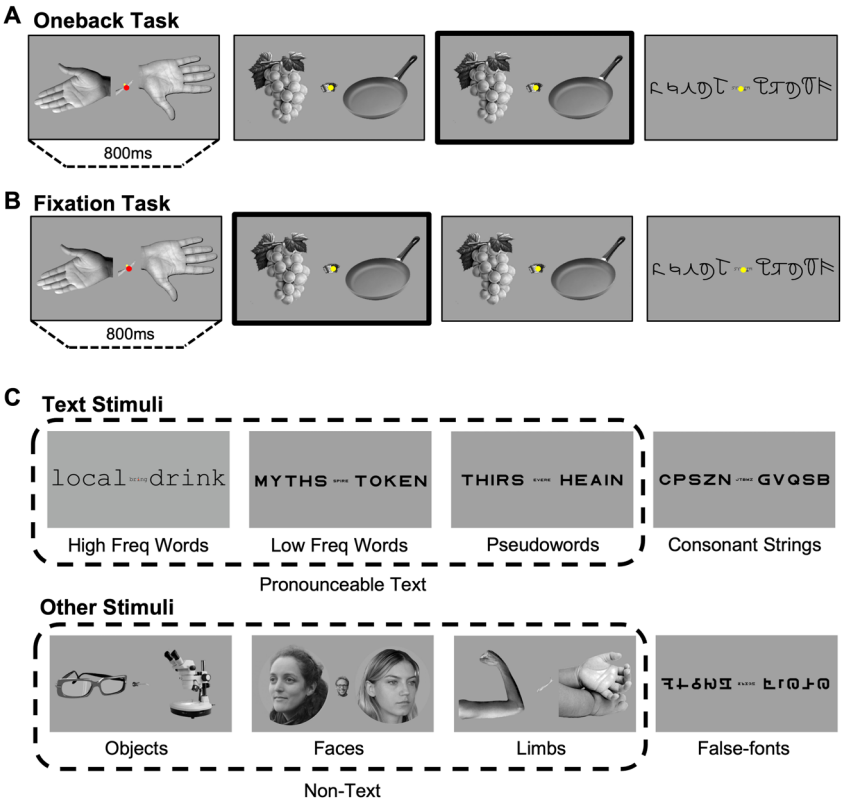


Figure 1. fMRI functional experiment. ***A***, One-back task: participants indicated via a button press when a stimulus appeared twice in a row. ***B***, Fixation task: participants indicated when a fixation dot at the center of the screen changed color. Stimuli were presented for 800 ms followed by 200 ms of a blank screen with a fixation dot. The frames where participants should respond in each task appear in bold. ***C***, Example stimuli grouped by visual categories. Text stimuli appear in the first row (high- and low-frequency words, pseudowords, and consonant strings), where only pronounceable text stimuli are circled. Other stimuli are shown in the bottom row (objects, faces, limbs, and false-fonts) with nontext stimuli circled. This separates the false-font category which shares some visual features with text. This illustration shows the faces of coauthors, but faces of people unfamiliar to participants were used in the actual experiment.

Functional runs were acquired with a gradient EPI sequence using a multiband factor of 3 to achieve whole-brain coverage (51 slices). We used a TR of 1.19 s with TE of 30 ms and a flip angle of 62 to obtain a spatial resolution of 2.4 mm^3 isotropic voxels. Each run included 232 frames and lasted 4:36 min. In addition, a high-resolution T1-weighted (T1w) anatomical scan was acquired with a spatial resolution of 0.9 mm^3 isotropic voxels.

Preprocessing. Preprocessing of functional data was performed using fMRIPrep 23.1.3 (Esteban et al., 2019), which is based on Nipype 1.8.6 (Gorgolewski et al., 2011, 2018).

Anatomical data preprocessing. The T1w image was reoriented into AC–PC alignment with ANTSpy version 0.4.2 (Tustison et al., 2021) and was used as an anatomical reference throughout the workflow. The T1w image was then skull-stripped using *Synthstrip* (Hoopes et al., 2022) and entered into FreeSurfer's recon-all pipeline (version 7.3.2; Dale et al., 1999) using *Synthseg* robust algorithm (Billot et al., 2023) for segmentation and surface reconstruction. All surfaces were visually inspected and edited manually in cases of inaccuracies. FreeSurfer derivatives were then used by the fMRIPrep pipeline. Brain tissue segmentation of cerebrospinal fluid (CSF), white matter (WM), and gray matter was performed on the brain-extracted T1w using *fast* (FSL; Zhang et al., 2001). Volume-based spatial normalization to one standard space (MNI152NLin2009cAsym) was performed through nonlinear registration with *antsRegistration* (Tustison et al., 2021), using brain-extracted versions of both T1w reference and the T1w template.

Functional data preprocessing. For each BOLD run, a reference volume was generated using a custom methodology of fMRIPrep. Head-motion parameters with respect to the BOLD reference (transformation matrices and six rotation and translation parameters) were estimated before any spatiotemporal filtering using *mcflirt* (FSL; Jenkinson et al., 2002). A B0-nonuniformity map was estimated based on two echo-planar imaging (EPI) references with *topup* (Andersson et al., 2003). The estimated field map was then aligned with rigid registration to the target EPI reference run. The field coefficients were mapped onto the reference EPI using the transform. BOLD runs were slice-time corrected to the middle slice using *3dTshift* from AFNI (Cox and Hyde, 1997). The BOLD reference was then coregistered to the T1w reference using *bbregister* (FreeSurfer) which implements boundary-based registration (Greve and Fischl, 2009). The time series of framewise motion parameters and the BOLD signal within the WM and the CSF were calculated and used later as confound factors in the BOLD response estimation. The BOLD time series were resampled into standard space, generating a preprocessed BOLD run in MNI152NLin2009cAsym space. The BOLD time series were also resampled onto the *fsnative* and *fsaverage* FreeSurfer surfaces. All resamplings were performed with a single interpolation step by composing all the pertinent transformations (i.e., head-motion transform matrices, susceptibility distortion correction, and coregistrations to anatomical and output spaces). Gridded (volumetric) resamplings were performed using *antsApplyTransforms* (ANTs), configured with Lanczos interpolation to minimize the smoothing effects of other kernels (Lanczos, 1964). Nongridded (surface) resamplings were performed using *mri_vol2surf* (FreeSurfer).

BOLD response estimation. We ran *mriqc* (version 22.0.1; Esteban et al., 2017) to inspect the quality of the data and exclude noisy runs. Runs where the mean framewise displacement (FD) was 0.5 mm or larger and runs where >30% of frames had FD >0.5 mm were excluded from the analysis. In addition, runs where participants failed to keep their eyes open were also excluded from the analysis. BOLD response estimation was performed by fitting a general linear model (GLM) using Nilearn (0.5.0; Nilearn contributors et al., 2024) to the BOLD time series in each participant's native surface. The design matrix for the GLM model included the white matter signal, the CSF signal, and their first derivatives, as well as the six motion parameters. In addition, first- and second-order polynomial drift parameters were included in the design matrix. The estimated BOLD responses (beta weights) are expressed in units of percent signal change (PSC) and reflect the change in response relative to the blank trials when the participants were presented with a blank fixation screen.

Regions of interest definition. We generated contrast maps comparing the estimated BOLD response to text stimuli (including high-frequency and low-frequency words, pseudowords, and consonant strings) with the responses to all other stimuli (false-fonts, faces, objects, limbs). Contrast maps were generated by weighing each visual category by the number of presented blocks to account for the higher number of text stimuli (20 blocks) compared with the other categories (10 blocks each). While we focus here on cross-sectional data from a single timepoint, to increase SNR for delineating ROIs, we created the contrast maps based on all available longitudinal data for each subject (6–20 runs, median = 15, collapsing across both tasks). Contrast maps were thresholded at a *t*-value of >2. Regions of interest (ROIs) were then defined for each individual in native space. The ROIs were defined with the following anatomical constraints which were defined based on each individual's anatomy: the first activation cluster was identified in the inferior frontal sulcus (IFS). This ROI was defined between the anterior border of the IFS or by a vertical boundary line if the sulcus extended into the curvature of the prefrontal cortex and the posterior border of the IFS. The IFS ROI was also constrained between the most dorsal sulcus of the IFS and the curvature of the inflated surface into the lateral sulcus (LS). The next clusters of activation were identified in the precentral gyrus (PrG). This ROI was defined between the anterior edge of the precentral sulcus and the anterior edge of the central sulcus. Ventrally, the PrG ROI was constrained by the curvature of the inflated surface into the lateral sulcus and dorsally by a horizontal boundary line approximately two-thirds of the way up from the ventral boundary. Visual inspection revealed that activations within the PrG tended to cluster in two separate areas; thus, we split this ROI into dorsal and ventral regions. To do this separation, a horizontal boundary line was defined on a group unthresholded probability map of the PrG. This boundary was projected into native space for each individual, effectively dividing their PrG ROI into dorsal and ventral components.

To investigate the relationship between reading skill and ROI size, a parallel set of ROIs was defined on contrast maps from the single time point (cross-sectional) data only, using the same definition criteria. This was done to reduce the potential variability in ROI size associated with the varying number of runs included in the GLM.

Statistical analysis

We first extracted the response magnitude for each stimulus category and task in units of percent signal change (PSC) from the baseline (blank fixation screen), averaged within each ROI. We used linear mixed effect (LME) models as implemented in the *lme4* package in R (Bates et al., 2015) to evaluate how activation in the frontal lobe ROIs is influenced by stimulus category and task demands. For each ROI, we fit a model with fixed effects of task (fixation vs one-back) and stimulus category, where we grouped the stimuli into four qualitatively different categories: pronounceable text (high-frequency and low-frequency words, pseudowords), consonant strings, false-fonts, and nontext (faces, objects, limbs). We selected these groupings based on the phonological properties of the stimuli, reflecting the IFC's role in the phonological processing of text. This approach was intended to highlight any potential differences in selective tuning to phonological features, compared with visual or orthographic features (sensitivity to letter shapes). All models included age and motion in the scanner as fixed effects, as well as random intercepts for each subject.

We next examined whether the ROIs exhibited similar response profiles. To this end, we fit a model with activation as the dependent variable and stimulus category and ROI as fixed variables [PSC ~ category × ROI + age + motion + (1 | subject)]. We used sum coding for the ROI variable to avoid treating any specific ROI as the baseline. Instead, each ROI was compared to the grand mean. In addition, we fit a similar model where the dependent variable was the magnitude of the task effect, defined as the subtraction of fixation activation from one-back activation [(One-back−fixation) ~ category × ROI + age + motion + (1 | subject)]. This allowed us to first examine the response properties across ROIs

over and above the different tasks and then tease apart whether the task effect differs between the ROIs. We followed each significant interaction with post hoc paired samples *t* tests.

We then investigated whether reading level modulates activation in the frontal lobe. To this end, we added *z*-scored individual reading scores [basic reading score (BRS) from the Woodcock–Johnson test] as a fixed effect to the models. For visualization, we split the sample into struggling readers and typical readers by a reading score cutoff of BRS <95. This cutoff equates to the 37th percentile, which is Woodcock–Johnson's cutoff for "average" readers. Note that this arbitrary cutoff was only used for visualization purposes, while in all statistical analyses, the BRS score was treated as a continuous variable. To follow up on the significant associations with reading, we fit post hoc multiple regression models for each task separately, predicting text selectivity (response to pronounceable text categories compared with all other categories) from reading skill, task performance (d'), age, and motion. Critically, task performance was included in these models to verify that any relationship observed with reading skill does not stem from differences in task performance.

We followed up on the ROI analysis with an exploratory vertex-wise analysis within a broad ROI encompassing the lateral frontal lobe, where we calculated the correlation between reading score and text selectivity (response to pronounceable text vs all other categories) within each vertex. This complementary analysis provides a more detailed picture of the spatial organization of the relationship between reading skill and text selectivity in the frontal lobe.

Lastly, to test whether reading skill was also associated with ROI size, we fit multiple linear regression models predicting ROI size (number of vertices) in the current timepoint from individual reading scores, age, and motion.

Results

Behaviorally, participants completed the one-back task with high accuracy [mean $d' = 1.75$, 95% CI: (1.54–1.95)] which was similar to their accuracy on the fixation task [mean $d' = 1.62$, 95% CI: (1.45–1.78); $t = 1.325$, $p = 0.189$]. Generally, participants responded faster in the one-back task with an average reaction time of 0.86 ± 0.14 s compared with the fixation task at 0.93 ± 0.15 s ($t = -4.78$, $p < 0.0001$). Together this suggests that the experimental design was successful in creating two tasks that were feasible for children and comparable in difficulty. The number of target trials per category precluded analysis of behavioral performance by category. Participants also tended to move more during the fixation task (mean FD = 0.21 mm) compared with the one-back task (mean FD = 0.19 mm; $t = 2.44$, $p = 0.018$). We therefore included motion per task (instead of average motion) in each statistical model that included task as one of the predictors. Reading score (measured outside the scanner) was weakly correlated with accuracy in the one-back task ($r = 0.262$, $p = 0.0339$) but not in the fixation task ($r = -0.027$, $p = 0.830$), and was not correlated with reaction time in either of the tasks (both $p > 0.1$).

At least three anatomically distinct text-selective regions in the frontal cortex

Text selectivity was defined based on the contrast comparing responses to text stimuli (high-frequency words, low-frequency words, pseudowords, consonant strings) with responses to all other categories (false-fonts, objects, faces, limbs). Visual inspection of these maps revealed at least three distinct activation peaks within each participant. We used anatomical boundaries on each individual's cortical surface to define these ROIs in order to maintain consistency across subjects (see Materials and Methods). Generally, the largest regions of activation were spread across the precentral gyrus (PrG) and clustered into dorsal and ventral regions. A more anterior patch (which was often diffuse and variable in size) was located along the inferior frontal sulcus (IFS). We identified and labeled these three text-selective ROIs within each participant by their anatomical location: IFS, dorsal PrG, and ventral PrG (Fig. 2). We were able to identify all three ROIs in all 66 subjects. Figure 2 shows the spatial organization of these ROIs on the native cortical surface of four individual subjects, as well as probability maps showing the ROI locations on the *fsaverage* template. When comparing the probability maps to FreeSurfer's anatomical atlases (Fischl et al., 2008), the IFS region partially overlapped Brodman's area 44 (pars opercularis) and 45 (pars triangularis), the ventral PrG region partially

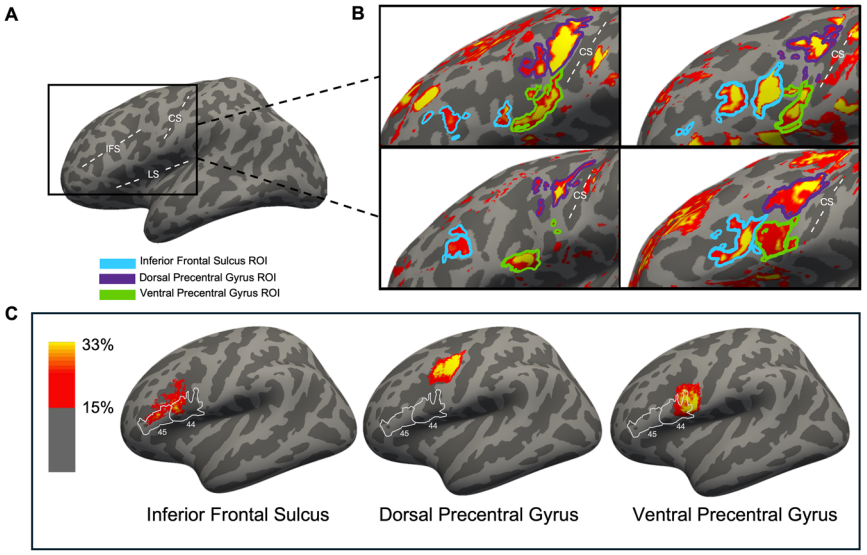


Figure 2. Anatomy of inferior frontal lobe text responses. ***A***, An example participant's left hemisphere inflated surface with anatomical labels: IFS, inferior frontal sulcus; CS, central sulcus; LS, lateral sulcus. ***B***, Four example subject's contrast maps, shown as heat maps (*t*-value >2), in native space with hand-drawn ROI labels: inferior frontal sulcus ROI in blue, dorsal precentral gyrus ROI in purple, and ventral precentral gyrus ROI in green. Note that additional patches of frontal lobe text activation exist, but we focused on the three regions that could be consistently defined in each individual. ***C***, Group average probability maps of ROI location in *fsaverage* space. Probability maps are thresholded at 15% overlap across subjects. FreeSurfer labels for Brodmann areas 44 and 45 (classical "Broca's region") are denoted by white contours as a reference.

overlapped area 44, and the dorsal PrG region was more dorsal than area 44. Thus, the dorsal PrG ROI does not seem to overlap at all with the canonical Broca's area. Importantly, even though the size and shape of these regions were variable across individuals, there was substantial overlap across the sample. However, in template space, very few vertices contained the functional region in more than one-third of the participants, highlighting the importance of defining ROIs on the individual's cortical surface.

Task enhances text activation in the frontal cortex

Across all ROIs, activation to text stimuli was significantly higher than activation for the other categories, which is expected as this was the definition criterion for the ROIs. In fact, there was hardly any activation to nontext stimuli (Fig. 3). Furthermore, the activation for consonant strings was weaker than the activation for pronounceable text in the IFS and dorsal PrG, suggesting some sensitivity to the pronounceability of the stimuli (Table 2). In all three ROIs, activation for pronounceable text during the one-back task was significantly greater than during the fixation task (Fig. 3 and Table 2), suggesting that a stimulus-directed task evokes a stronger neural response in these regions. Importantly, this task effect was highly specific for the pronounceable text stimuli, as indicated by a significant interaction between task and stimulus category (Table 2 and Fig. 3). Specifically, the task effect was stronger in the pronounceable text category compared with the nontext category in all ROIs. This suggests that these regions are modulated by cognitive processes specific to text, as opposed to more domain-general mechanisms which would enhance activation of all attended visual stimuli. Interestingly, for the IFS alone, the task effect for pronounceable text was also significantly stronger than the task effect for false-fonts, suggesting that the IFS is sensitive to real letter shapes, as opposed to visually matched false-fonts. In the dorsal PrG alone, activation was negatively associated with motion per task.

To directly compare the ROIs, we ran an additional LME where we evaluated whether the response magnitude was affected by the interaction between stimulus category and ROI (see Materials and Methods for model formula). There was no main effect of ROI ($p > 0.1$), indicating that all ROIs showed similar response magnitudes overall. There was also no interaction

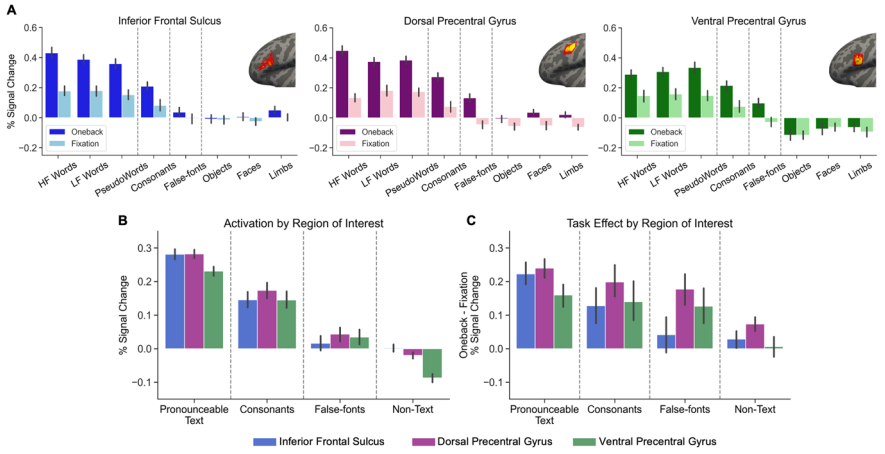


Figure 3. Individual ROI activation for each visual stimulus and task. ***A***, Activation is shown as the mean percent signal change from baseline for each visual category and for each task (one-back and fixation). To represent the within-subject variability, we subtracted each subject's mean PSC from each datapoint and added the group mean. Error bars represent the 68% confidence interval. ***B***, A comparison of the average activation between ROIs. Visual stimuli were grouped into categories (pronounceable text, consonant strings, false-fonts, and nontext) for clarity. ***C***, The magnitude of task effect, calculated as the average difference between one-back and fixation PSC values within each subject.

Table 2. Linear mixed effect models examining the relationship between stimulus category (pronounceable text, consonant strings, false-fonts, and nontext) and task (one-back vs fixation)

	Activation (PSC) ~ category × task + age + motion + (1 | subject)								
	IFS			Dorsal PrG			Ventral PrG		
	β	*t*	*p*	*β*	*t*	*p*	*β*	*t*	*p*
(Intercept)	−0.125	−0.570	0.5701	0.292	1.294	0.1997	0.129	0.539	0.5917
Category									
Consonants	−0.088	−2.413	0.0160*	−0.088	−2.738	0.0063*	−0.076	−1.938	0.0529
False-fonts	−0.174	−4.770	0.0000*	−0.208	−6.458	0.0000*	−0.180	−4.602	0.0000*
Nontext	−0.182	−7.035	0.0000*	−0.219	−9.641	0.0000*	−0.241	−8.737	0.0000*
Task									
One-back	0.224	8.563	0.0000*	0.229	9.954	0.0000*	0.157	5.614	0.0000*
Category × task									
Consonants: One-back	−0.095	−1.838	0.0663	−0.041	−0.906	0.3650	−0.020	−0.361	0.7179
False-fonts: One-back	−0.182	−3.509	0.0005*	−0.063	−1.375	0.1696	−0.033	−0.602	0.5474
Nontext: One-back	−0.194	−5.309	0.0000*	−0.166	−5.171	0.0000*	−0.153	−3.931	0.0001*
Age	0.028	1.373	0.1742	−0.001	−0.062	0.9511	0.006	0.246	0.8062
Motion	0.065	0.319	0.7500	−0.553	−2.908	0.0038*	−0.157	−0.712	0.4771

Age and in-scanner motion per task (mean framewise displacement in millimeters) were entered as fixed effects of no interest. Nontext included faces, limbs, and objects. Pronounceable text was defined as the reference level for the category variable, and fixation was the reference level for the task variable. The asterisks denote a significance level of $p < 0.05$.

between ROI and stimulus category ($p > 0.1$), suggesting that the three regions show similar degrees of selectivity to text (Fig. 3*B*). We next ran a similar LME predicting the magnitude of the task effect (as defined by the subtraction of activation during fixation from the activation during one-back). There was no main effect of ROI on task effect magnitude ($p > 0.05$), indicating that the task effect was similar in strength across ROIs. Interestingly, we also observed a significant interaction between ROI and stimulus category [Fig. 3*C*; category (false-fonts) × ROI (IFS), $\beta = -0.089$, $p = 0.0175$]. Post hoc analyses revealed that the task effect was significantly stronger for pronounceable text compared with false-fonts in the IFS (paired t test 4.597, $p = 0.0002$) while this was not the case in the dorsal PrG ($t = 1.795$, $p = 0.077$) or ventral PrG ($t = 0.891$, $p = 0.376$). This suggests that the IFS is sensitive to the text similarity of the stimuli more than the PrG ROIs. This was complemented by the PrG ROIs, which showed a significantly stronger task effect for false-fonts compared with nontext ($t = 3.246$, $p = 0.0019$; $t = 3.442$, $p = 0.001$; dorsal and ventral, respectively), while the IFS did not ($t = 0.32$, $p = 0.75$). This highlights the difference in response properties between the IFS and PrG ROIs.

Lastly, to further tease apart whether any of the ROIs were specifically sensitive to lexical information versus phonological information, we ran a set of post hoc paired samples t test, comparing the response to lexical stimuli (high- and low-frequency words) with the response to pseudowords. Paired samples t tests revealed no lexicality effect in either of the ROIs, for either of the tasks (all $p > 0.1$). However, all ROIs showed increased activation to pseudowords compared with consonant strings during the one-back task (IFS, $t = 4.14$, $p = 0.0001$; dorsal PrG, $t = 3.87$, $p = 0.0003$; ventral PrG, $t = 3.51$, $p = 0.0008$). During the fixation task, only the dorsal PrG still showed a preference to pseudowords (IFS, $t = 1.58$, $p = 0.119$; dorsal PrG, $t = 2.90$, $p = 0.005$; ventral PrG, $t = 1.66$, $p = 0.102$). In contrast, the magnitude of the task effect did not significantly differ across any of the text categories (all t tests $p > 0.1$). Taken together, these results suggest that while these ROIs are sensitive to text pronounceability, the task-specific enhancement of responses to text applies to any textual stimulus regardless of its pronounceability or lexical content and is therefore more likely visually driven.

Reading skill modulates response to text in the frontal cortex

We next tested whether activation in each ROI was associated with reading skill. Combining across tasks, better reading skill was associated with stronger activation to pronounceable text, especially in the dorsal PrG (Table 3, Fig. 4). In both the IFS and the dorsal PrG, the association between reading skill and activation strength was significantly stronger for pronounceable text stimuli, compared with nontext stimuli (Table 3). The dorsal PrG showed further specificity, as the association with reading was significantly stronger for pronounceable text even when compared with false-fonts and consonant strings. This suggests that reading proficiency is specifically associated with greater responses to pronounceable text in these regions, rather than

Table 3. Linear mixed effect models examining the relationship between reading skill and activation (top) or task effect (bottom)

	Activation (PSC) ∼ category × reading + age + motion + (1 | subject)								
	IFS			Dorsal PrG			Ventral PrG		
	β	*t*	*p*	*β*	*t*	*p*	*β*	*t*	*p*
(Intercept)	−0.103	−0.422	0.6747	0.396	1.591	0.1167	0.029	0.110	0.9126
Category									
Consonants	−0.136	−5.062	0.0000*	−0.109	−4.435	0.0000*	−0.086	−3.030	0.0025*
False-fonts	−0.265	−9.888	0.0000*	−0.239	−9.752	0.0000*	−0.196	−6.944	0.0000*
Nontext	−0.279	−14.712	0.0000*	−0.302	−17.453	0.0000*	−0.318	−15.908	0.0000*
Reading (BRS *z*-score)	0.058	1.976	0.0516	0.092	3.110	0.0026*	0.062	1.989	0.0500
Category × reading									
Consonants: reading	−0.046	−1.707	0.0882	−0.069	−2.797	0.0053*	0.000	0.004	0.9967
False-fonts: reading	−0.039	−1.470	0.1418	−0.054	−2.205	0.0277*	−0.010	−0.359	0.7199
Nontext : reading	−0.048	−2.530	0.0116*	−0.037	−2.160	0.0310*	−0.020	−0.981	0.3269
Age	0.032	1.510	0.1361	−0.002	−0.083	0.9339	0.013	0.585	0.5609
Motion	0.325	0.923	0.3595	−0.478	−1.330	0.1884	0.351	0.937	0.3525
	Task effect ∼ category × reading + age + motion + (1 | subject)								
	IFS			Dorsal PrG			Ventral PrG		
	β	*t*	*p*	*β*	*t*	*p*	*β*	*t*	*p*
(Intercept)	−0.203	−0.558	0.5787	−0.033	−0.110	0.9128	−0.060	−0.145	0.8853
Category									
Consonants	−0.095	−2.517	0.0122*	−0.041	−1.179	0.2392	−0.020	−0.512	0.6088
False-fonts	−0.182	−4.804	0.0000*	−0.063	−1.788	0.0745	−0.033	−0.853	0.3941
Nontext	−0.194	−7.269	0.0000*	−0.166	−6.725	0.0000*	−0.153	−5.572	0.0000*
Reading (BRS *z*-score)	0.004	0.089	0.9291	0.036	0.974	0.3328	−0.049	−0.989	0.3259
Category × reading									
Consonants: reading	0.015	0.386	0.6997	−0.005	−0.130	0.8968	0.043	1.093	0.2752
False-fonts: reading	0.015	0.400	0.6891	0.048	1.369	0.1718	0.027	0.691	0.4901
Nontext : reading	−0.035	−1.326	0.1855	0.016	0.651	0.5157	0.064	2.326	0.0204*
Age	0.034	1.064	0.2913	0.027	1.051	0.2972	0.026	0.720	0.4740
Motion	0.451	0.857	0.3946	−0.006	−0.014	0.9886	−0.197	−0.328	0.7437

Task effect was defined as the difference between the one-back task and the fixation task. Reading was entered into the model as the *z*-scored basic reading score (BRS) from the Woodcock–Johnson reading test. Age and in-scanner motion (mean framewise displacement in mm) were entered as fixed effects of no interest. Pronounceable text was defined as the reference level for the category variable. The asterisks denote a significance level of $p < 0.05$.

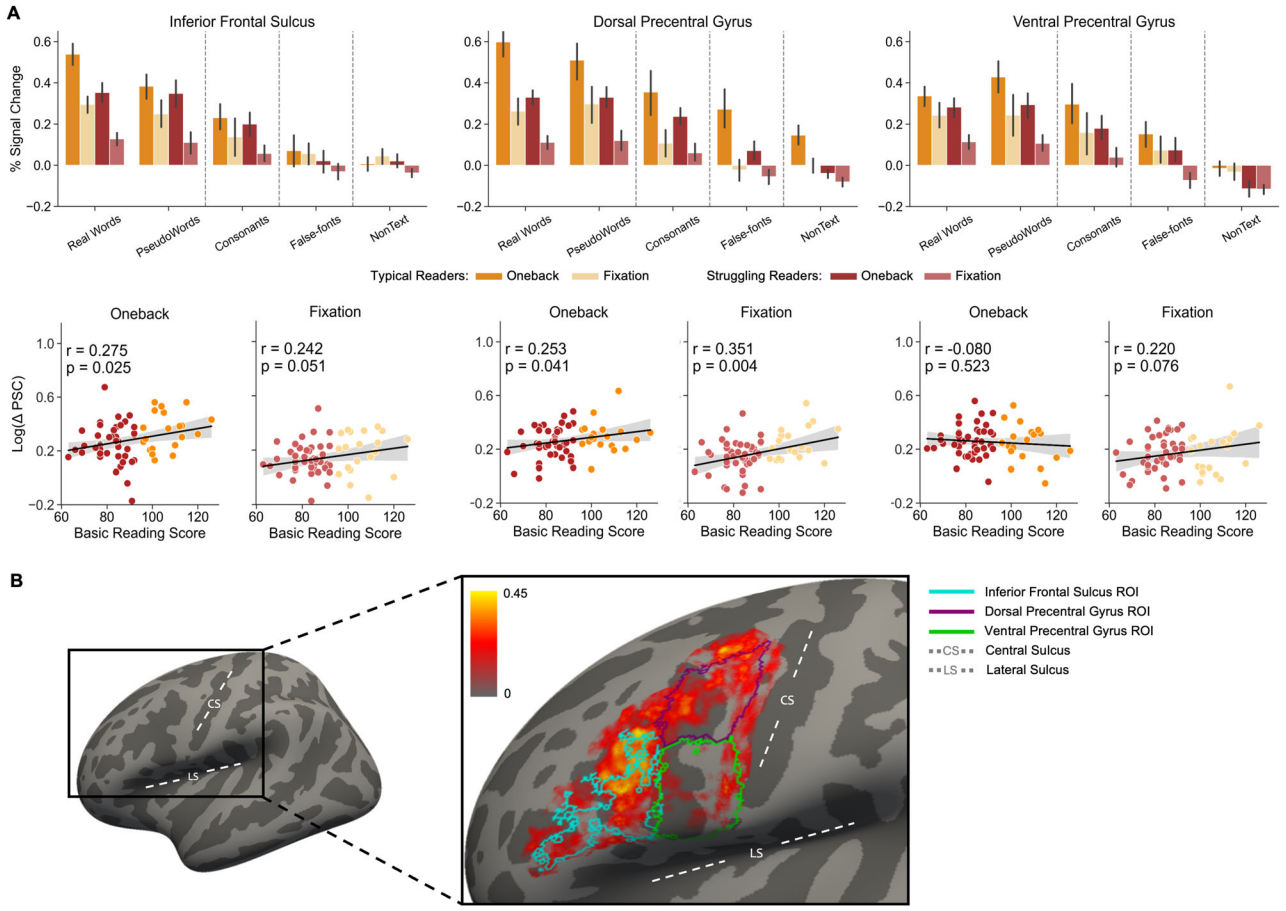


Figure 4. Relationship between activation in text-selective cortex and reading scores. ***A***, Average activation for each visual category, split by task and by reading level (based on an arbitrary cutoff for visualization purposes). Typical readers are indicated in orange ($N = 20$) and struggling readers in red ($N = 46$). The one-back task is indicated by the darker color and the fixation task by the lighter color within each reading group. Error bars represent a 68% confidence interval. Scatter plots underneath each ROI display the Pearson's correlation between WJ basic reading score and activation for pronounceable text—all other visual categories (log of Δ PSC), for each task. ***B***, Spatial distribution of the relationship between reading skill and text selectivity. We transformed each participant's *z*-scored contrast map from native surface to FreeSurfer's *fsaverage* template and calculated vertex-wise Pearson's correlation between individual selectivity to text and WJ basic reading score. Shown are the correlation coefficients for each vertex mapped onto an inflated *fsaverage* surface for visualization, overlayed with contours of the group average ROIs based on the group probability map (Fig. 2).

an overall pattern of elevated responses across visual categories. When focusing on the task effect as the predicted variable, reading skill was not associated with the magnitude of the task effect for text stimuli in any of the ROIs. In the ventral PrG alone, there was a significant interaction such that reading skill was associated with a larger task effect in the nontext category.

We followed up on the significant associations with reading by calculating post hoc Pearson's correlations between reading skill and text selectivity within each task separately (response to pronounceable text categories compared with all other categories; Fig. 4). This analysis showed that reading skill was positively correlated with text selectivity in the IFS (one-back $r = 0.275$, $p = 0.025$; fixation $r = 0.242$, $p = 0.051$) and the dorsal PrG (one-back $r = 0.253$, $p = 0.041$; fixation $r = 0.351$, $p = 0.004$) but less so in the ventral PrG (one-back $r = -0.080$, $p = 0.523$; fixation $r = 0.220$, $p = 0.076$). To take into account task performance, we repeated these analyses by fitting multiple regression models in each ROI, predicting text selectivity in each task from reading scores, task performance (d'), age, and motion. This revealed that reading skill was a significant predictor of text selectivity in the IFS during the one-back task ($\beta = 0.054$, $t = 2.226$, $p = 0.030$) and in the dorsal PrG during the fixation task ($\beta = 0.055$, $t = 3.011$, $p = 0.004$) and to a lesser extent in the ventral PrG ($\beta = 0.044$, $t = 2.013$, $p = 0.049$). The rest of the predictor variables did not make a significant contribution to the models ($p > 0.05$) except for age, which was negatively associated with text selectivity during the fixation task in the ventral PrG alone ($\beta = -0.039$, $t = -2.315$, $p = 0.024$). This suggests that older children showed weaker text selectivity in this specific ROI. To follow up on this finding, we also calculated the simple Pearson's correlation between age and text selectivity in each task and confirmed that the only significant correlation was found in the ventral PrG, where age was negatively associated with text selectivity during the fixation task ($r = -0.286$, $p = 0.020$). Importantly, task performance (d') was not a significant predictor of text selectivity in any of the regression models.

To further investigate the spatial distribution of the relationship between reading skill and text selectivity in the frontal lobe, we calculated the vertex-wise Pearson's correlation between reading scores and text selectivity within a broad frontal lobe ROI. The resulting map (Fig. 4*B*) shows a high spatial overlap between the peak correlations and the dorsal parts of the IFS and dorsal PrG ROIs. This map shows a clear dissociation between the dorsal PrG and the ventral PrG, where the effect of reading skill is almost absent. This supports the notion that these ROIs are distinct and have different relationships with reading.

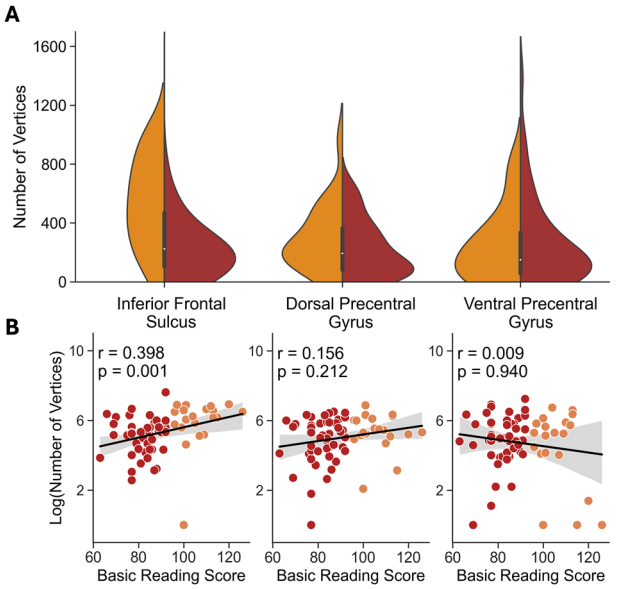


Figure 5. Reading skill is associated with ROI size. ***A***, Distribution of ROI size in number of vertices for typical readers (orange, $N = 20$) and struggling readers (red, $N = 46$). ***B***, Scatterplots display the Pearson's correlation between the log number of vertices and the WJ basic reading score for each ROI. Typical readers are indicated in orange and struggling readers in red.

Finally, we tested whether reading skill was associated with ROI size. We found that struggling readers had smaller IFS ROIs ($\beta = 9.892$, $p = 0.001$, Fig. 5). However, the size of the dorsal PrG and ventral PrG ROIs was not associated with reading ability ($\beta = 2.111$, $p > 0.1$, and $\beta = 1.145$, $p > 0.1$, respectively). This again highlights the importance of the IFS region in reading and the variability in the size of the region across participants. Age and motion were included in the multiple regression models and were not associated with the size of any of the ROIs (all $p > 0.1$). We also didn't observe any direct correlation between age and ROI size in any of the ROIs, confirming that the correlation with reading ability was unlikely confounded by age.

Discussion

Our findings indicate that three anatomically distinct, text-selective regions can be identified in the inferior frontal cortex of young readers. These regions show a considerable effect of the cognitive task that the participant is engaged in, and this task effect is highly specific to text stimuli. Additionally, we found that individual differences in reading ability are related to activation levels in these regions, such that better readers show stronger activation for text stimuli, and larger clusters of text-selective responses in the IFS. Although all three regions were identified in the same way (based on their selectivity to text), our findings reveal distinct functional profiles. In the IFS alone, the task effect differentiates between false-fonts and text stimuli, suggesting a task-driven enhancement that is highly selective to stimulus orthography. The dorsal PrG shows the most specific and robust relationship with reading skill, while the ventral PrG does not show a clear relationship with reading skill.

A growing number of studies have recognized that language regions in IFC comprise multiple, functionally distinct subregions (Fedorenko et al., 2012; Fedorenko and Blank, 2020; Wang et al., 2021, 2023; Chauhan et al., 2024). Some distinguish between a phonological subregion, typically more posterior and dorsal and a semantic subregion which is more anterior and ventral (Poldrack et al., 1999; Vigneau et al., 2006; Wang et al., 2021), in line with the dual-route model of language (Hickok and Poeppel, 2007; Saur et al., 2008). Specifically in IFC, it has been suggested that the pars opercularis of "Broca's area" plays a part in the dorsal, phonological stream, while pars triangularis is part of the ventral, semantic stream (Saur et al., 2008; Skeide et al., 2016). Others have noted that the broadly defined "Broca's area" comprises a language subregion and a domain-general subregion that are spatially distinct (Fedorenko et al., 2012; Fedorenko and Blank, 2020). In the current study, we identified three regions in IFC that are all highly selective to text, yet differ on several functional aspects. When considering the task effect, the more anterior IFS region shows the highest sensitivity to text visual features. In contrast, the dorsal and ventral PrG regions show a similar task effect in the false-font category. Additionally, both the IFS and the dorsal PrG show a strong relationship with reading skill, while this is not the case for the ventral PrG. The distinction between these ROIs was further confirmed by a vertex-wise analysis, where peaks of correlation between text selectivity and reading skill overlapped with the IFS and the dorsal PrG, but not with the ventral PrG. This suggests that the anatomical distinctions we drew represent true functional differences between subregions of IFC. Interestingly, we found that all ROIs were sensitive to stimulus pronounceability, but not to lexical status. This supports a view of these regions as pertaining to the dorsal, phonological reading stream. Of note, these ROIs are likely more dorsal than the portion of "Broca's area" where lexical and semantic effects are typically observed.

There is a growing understanding that task choice plays a key role in neuroimaging and has a critical influence on measured responses (Kay et al., 2023). A common practice to isolate language regions from other functional regions is to engage participants in a language task and compare it to a control task that does not recruit the language system. This approach manipulates both the stimulus category (e.g., words vs faces) and the task (e.g., lexical decision vs visual task) at the same time (Fedorenko and Blank, 2020; Borghesani et al., 2021; White et al., 2023). These comparisons are useful in identifying language regions but conflate the stimulus-driven response with the cognitive task and the mental operations the participants perform. Thus to properly examine task effects in the text-selective cortex, it is crucial to hold the visual stimuli constant while varying the task and to hold the task constant while varying the visual stimulus. This approach showed that different tasks can enhance or attenuate the responses to different stimuli in the category-selective cortex (Kay et al., 2015; Kay and Yeatman, 2017; White et al., 2023). Interestingly, studies that examined task effects on a constant set of stimuli have found that patterns of neural representation in IFC were strongly task-dependent, to a greater extent than in the ventral occipitotemporal cortex (Johnston and Everling, 2006; McKee et al., 2014; Bugatus et al., 2017; Jung and Walther, 2021). Furthermore, McKee et al., (2014) found that in primates, tasks that recruited attention enhanced category selectivity in IFC. Our results show that the one-back task, which directs attention toward the stimulus, similarly enhances the response in a category-specific manner (i.e., for text stimuli). Along these lines, a recent study reported an interaction between task and stimulus in two IFC subregions in adults (Chauhan et al., 2024). Our findings expand this observation by characterizing the spatial organization of text-selective regions in individual children and uncovering their relationship with reading skill.

Another key point regarding task choice is our decision to use tasks that do not necessarily require language processing. Previous studies have typically located language regions using tasks that explicitly require language processing, like sentence or word reading, naming, lexical decision, or auditory phonological processing (Booth et al., 2007; Hoeft et al., 2007; Maisog et al., 2008; Fedorenko et al., 2012; Kovelman et al., 2012; Olulade et al., 2015; Wang et al., 2021, 2023; Zhang and Peng, 2022). When studying the relationship between brain activation and reading ability, this approach can intermingle individual differences with task difficulty: for example, in an extreme case, a participant who can't read will not be able to complete a reading task and activation differences will be meaningless. This will inevitably reveal group differences between children who struggle with the task itself and children who do not. Discrepancies between past studies reporting reduced or elevated IFC responses in struggling readers could stem from employing tasks with varying levels of reliance on phonological knowledge, reading, or language skills that are impaired in dyslexia. Here, we took special care to compare two tasks that can, in principle, be performed without relying on linguistic knowledge and can be done on any type of stimulus (either textual or not). Although we cannot fully rule out that some automated decoding of the text stimuli takes place, it is not required in order to perform the tasks well. While higher-level N-back tasks are typically considered a demanding measure of working memory, the one-back version incurs relatively low cognitive load as it only requires holding the most recent item in memory without suppressing responses to intervening stimuli (Pelegrina et al., 2015). Indeed, we found that participants responded faster on the one-back task compared with the fixation task, and reading ability was not correlated with reaction times for either task. While reading ability was weakly correlated with one-back accuracy, our post hoc analyses confirmed that this had no effect on the relationship between reading ability and text selectivity of the neural responses. Our results therefore show that more skilled readers demonstrate higher tuning to visually presented text in IFC. Importantly, we find this effect even in the fixation task that directs attention away from the stimuli.

The task effect we observed cannot be explained as a domain-general, attention-driven amplification, and, thus, we are careful not to use the term "attention" to describe the task effects reported here. Such a mechanism would result in increased activation in the one-back task for all visual stimuli, as studies have reported in the ventral occipitotemporal cortex (Kay and Yeatman, 2017), whereas we only see task-driven increased activation specific to text stimuli. Additionally, we verify these regions are only selective for text rather than domain-general in three ways: (1) the regions were defined on text-selective contrasts of text > other visual stimuli, and (2) we see hardly any activation for nontext categories (objects, faces, limbs), as opposed to category-selective regions in the occipitotemporal cortex that also respond to other visual categories to different degrees (Ben-Shachar et al., 2007; Centanni et al., 2017; Kay and Yeatman, 2017; Kubota et al., 2019; White et al., 2023; Dalski et al., 2024). (3) The task effect is absent for nontext categories and differentiates between text and false-fonts in the case of the IFS. Therefore, we conclude that this task effect, or cognitive enhancement, is specific to text.

To conclude, we identified three anatomically distinct text-selective regions in IFC which are highly selective to text even during tasks that do not explicitly require reading. We found that activation in these regions is associated with reading proficiency such that better readers show higher activation to text stimuli and greater text selectivity. To our knowledge, this is the first study demonstrating that skilled readers have an increased response to purely visual text in the frontal cortex. However, it remains unclear how and when in the process of reading development this relationship comes about. The current cross-sectional data do not allow us to distinguish between two alternative hypotheses. It could be that accumulating reading experience tunes the responses in the frontal lobe to be more selective to text. Alternatively, it could be that children who develop more selective responses in the frontal cortex are the ones who will master reading more easily. It also remains an open question whether similar anatomical distinctions can be described in the right frontal cortex and to what extent text-selective task effects are lateralized. Future studies employing longitudinal methodologies could determine the existence and direction of a causal relationship between these regions and reading skill.

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